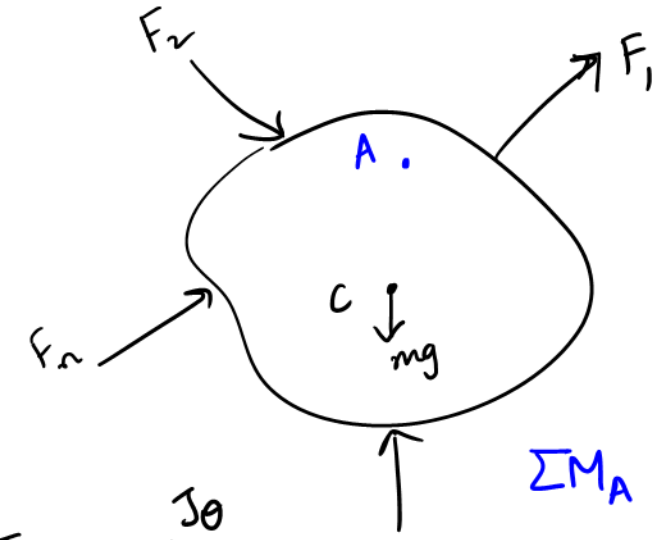
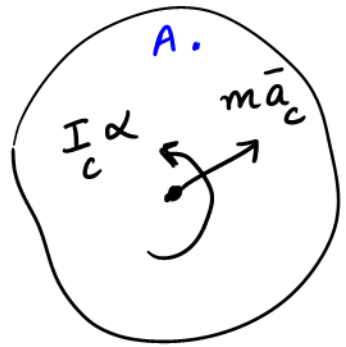


ΣM_A

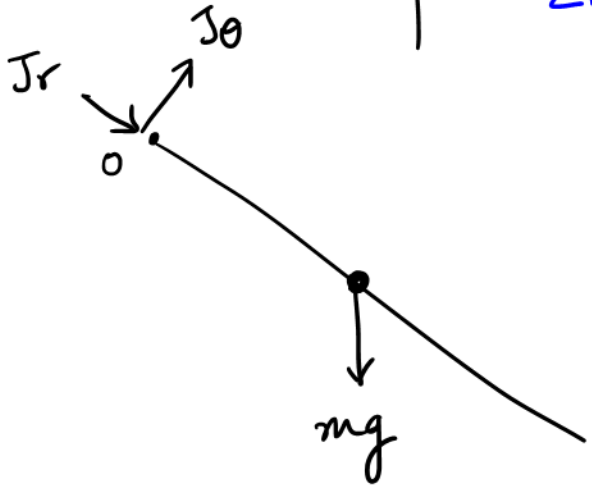
Kinetic diagram



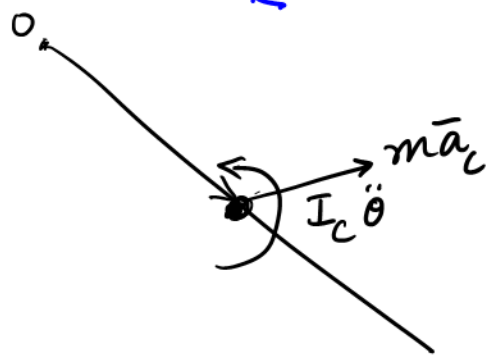
$\Sigma \bar{F} = m\bar{a}_c$

$\Sigma M_c = I_c \alpha$

$= (\Sigma M_A)_K$



$=$



$\bar{a}_c = l\ddot{\theta}\,\hat{e}_\theta - l\dot{\theta}^2\hat{e}_r$

$I_c = \frac{m}{12}(2l)^2 = \frac{ml^2}{3}$

$\bar{r}_c = r_c\hat{e}_r = l\hat{e}_r$

$\Sigma M_o = I_c\ddot{\theta} + r_c\hat{e}_r \times m\bar{a}_c = I_c\ddot{\theta} + l\hat{e}_r \times m(l\ddot{\theta}\hat{e}_\theta - l\dot{\theta}^2\hat{e}_r)$

$= \frac{ml^2}{3}\ddot{\theta} + ml^2\ddot{\theta} = \frac{4}{3}ml^2\ddot{\theta}$

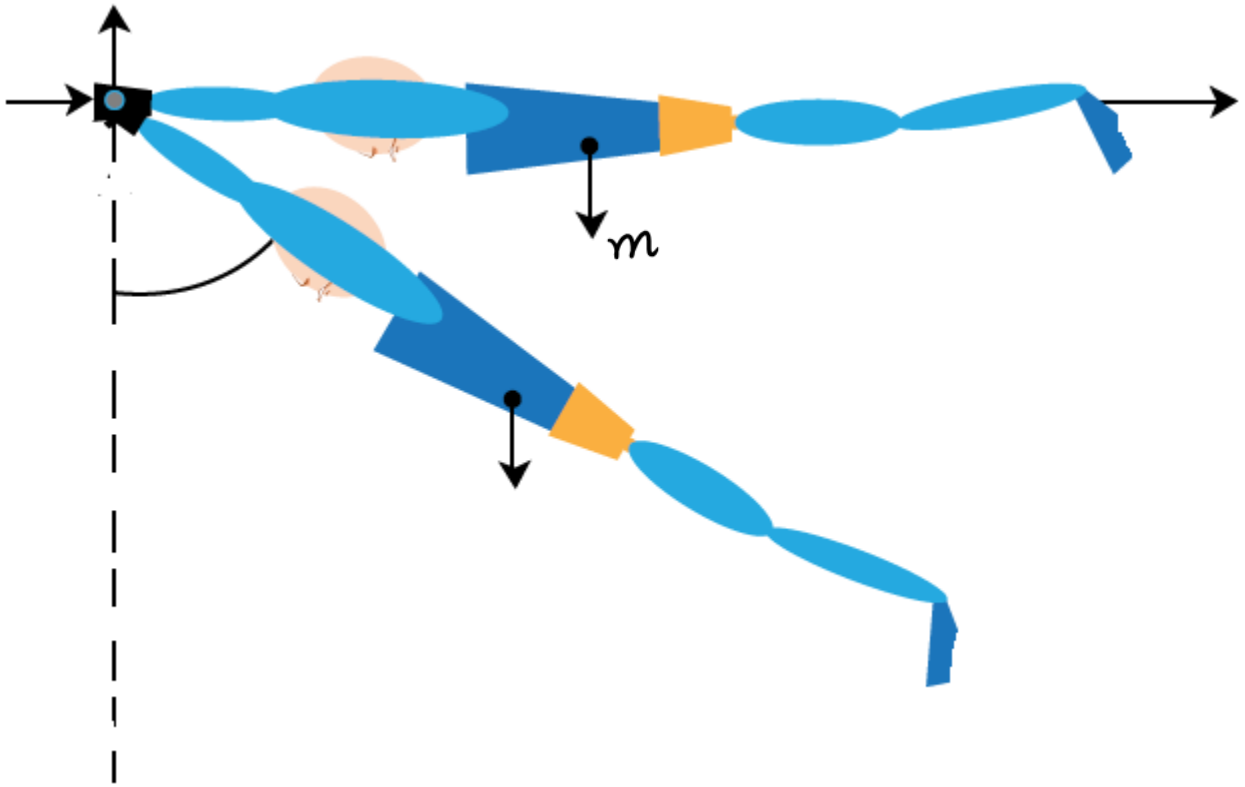
$\boxed{\Sigma M_o = I_o\alpha}$

$I_c \left[\frac{ml^2}{3} + ml^2\right]$ $\parallel$ axis theorem

$I_o = I_c + ml^2$

Parallel-axis theorem

Angular motion – concentrated mass



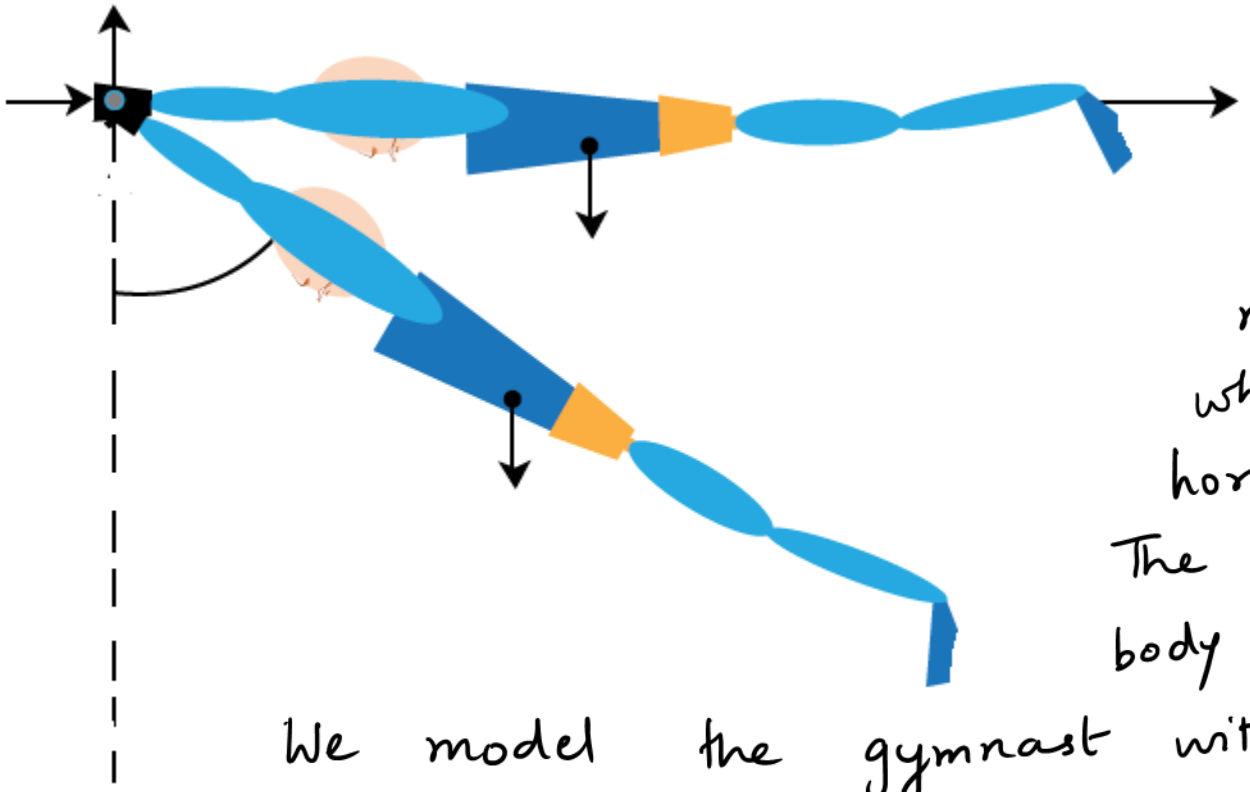
$$\Sigma \bar{F} = m \bar{a}_c$$

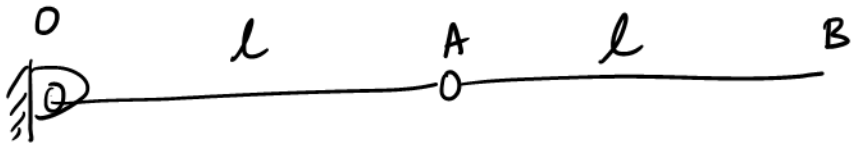
$$I_o = m l^2$$

With a uniformly distributed mass

$$I_o = \frac{4}{3} m l^2$$

Angular motion – distributed mass, two-rod model





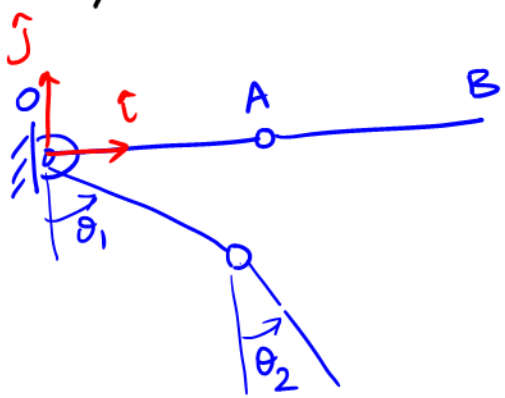
We want to assess the loads carried by the back/abdominal muscles of the gymnast of mass 2m when the gymnast is let go from a horizontal position.

The gymnast swings down keeping her body aligned in a st. line

We model the gymnast with two rods OA & AB connected by a hinge at pt. A.

When the two rods are let go from rest in the horizontal configuration, will they begin to rotate as one solid body?
What muscles (back/abs) must the gymnast contract to remain straight?

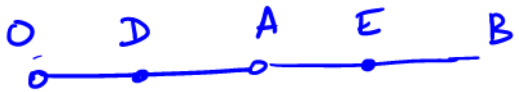
Ref:
Aydin Tozeren : Dynamics of the Human Body



Released from rest $\quad \dot{\theta}_1 = \omega_1 = 0, \quad \dot{\theta}_2 = \omega_2 = 0, \quad \theta_1 = \theta_2 = \pi/2$

$$\bar{a}_D = \alpha_1 \frac{l}{2} \hat{j}$$

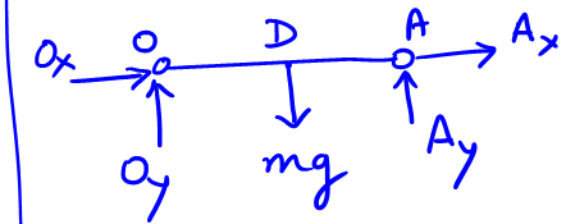
$$\bar{a}_E = \bar{a}_A + \bar{a}_{E/A}$$
$$= \alpha_1 l \hat{j} + \alpha_2 \frac{l}{2} \hat{j}$$



2 pts in a rigid body



$$\bar{v}_Q = \bar{v}_P + \bar{\omega} \times \bar{r}_{Q/P}$$
$$\bar{a}_Q = \bar{a}_P + \bar{\alpha} \times \bar{r}_{Q/P} + \bar{\omega} \times (\bar{\omega} \times \bar{r}_{Q/P})$$



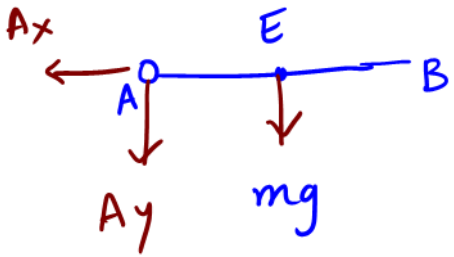
$$O_x + A_x = -m\omega_1^2 \frac{l}{2} = 0$$

$$O_y - mg + A_y = m\alpha_1 \frac{l}{2}$$

$$\Sigma M_O = I_O \alpha_1 \quad \circlearrowleft +$$

$$-mg\frac{l}{2} + A_y l = \frac{ml^2}{3}\alpha_1$$

$\frac{ml^2}{3}$ → moment of inertia of rod OA about fixed pt. O



$\sum M_E = I_E \alpha_2$

$\circlearrowleft +$

$A_x = 0$ ∵ $\omega_2 = 0$ at this instant

$-A_y - mg = m a_{E_y} = m\left(\alpha_2 \frac{l}{2} + \alpha_1 l\right)$

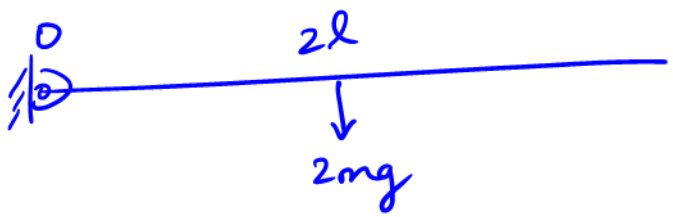
$A_y \frac{l}{2} = \frac{ml^2}{12} \alpha_2$

$\therefore A_y = \frac{ml}{6} \alpha_2$

Six unknowns : $O_x, O_y, A_x, A_y, \alpha_1, \alpha_2$

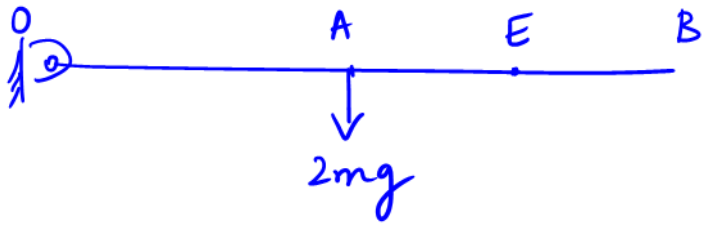
Solving, $\alpha_1 = \frac{-9g}{7l}$, $\alpha_2 = \frac{3g}{7l}$

If the body remains straight from the beginning

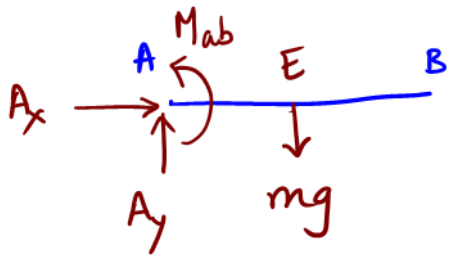


$\sum M_O = I_O \alpha$

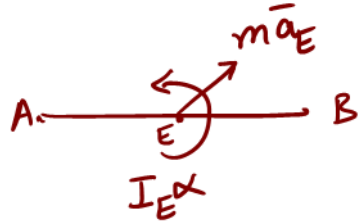
$-2mgl = \frac{(2m)(2l)^2}{3} \alpha \Rightarrow \alpha = \frac{-3g}{4l}$



$$a_E = \alpha \cdot \frac{3\ell}{2} = -\frac{3}{4}\frac{g}{\ell} \cdot \frac{3\ell}{2} = -\frac{9}{8}g$$



=



E is the CoM of this rod.

$$\sum M_A = I_E\alpha + \bar{r}_{E/A} \times m\bar{a}_E$$

$$= \frac{m\ell^2}{12}\left(\frac{-3g}{4\ell}\right) + \frac{\ell}{2}\hat{i} \times \left(-\frac{9}{8}mg\right)\hat{j}$$

$$= -\frac{mg\ell}{16} - \frac{9\,mg\ell}{16} = -\frac{5}{8}mg\ell$$

$$\sum M_A = -mg\frac{\ell}{2} + M_{ab} = -\frac{5}{8}mg\ell$$

$$M_{ab} = -\frac{5}{8}mg\ell + \frac{1}{2}mg\ell = \frac{-mg\ell}{8}$$

abdominal muscles agonists

Net effect of agonist & antagonist muscles